

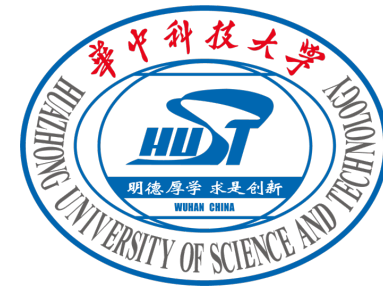
人工智能原理

元宇宙导论

电子信息与通信学院
廖振宇



个人介绍



- **教育和工作经历:**

- 2010-2014, 本科, 光电信息工程, 华中科技大学
- 2014-2016, 硕士, 信号与图像处理, 法国巴黎萨克雷大学
- 2016-2019, 博士, 计算机与数学, 法国巴黎萨克雷大学
- 2019-2020, 博士后研究员, 统计系 & ICSI, 美国加州大学伯克利分校
- 2021-至今, 副研究员, 电子信息与通信学院, 华中科技大学

- **个人荣誉:**

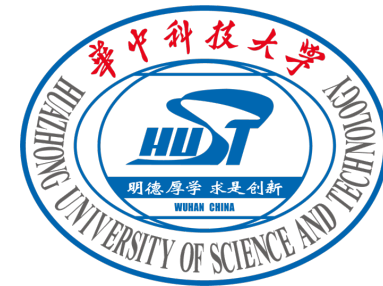
- 第十四批湖北省“百人计划”（创新人才）、“武汉英才”优秀青年人才、华中科技大学东湖青年学者
- 法国自然科学基金 ANR-LabEx-CIMI 访问教授、加拿大 CRM-Simons 访问教授等

- **学术服务:**

- 欧盟自然科学基金 (ERC)、加拿大国家自然科学基金 (NSERC) 和以色列科学基金会 (ISF) 评审
- 机器学习旗舰会议 ICML、ICLR、IJCNN 领域主席 (Area Chair)
- Springer Statistics and Computing 副编辑 (Associate Editor)
- 中国现场统计研究会随机矩阵理论与应用分会副秘书长, 中国现场统计研究会大数据统计分会理事

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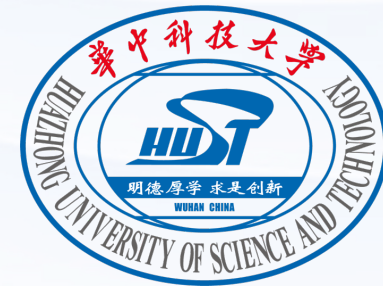
1. 人工智能和机器学习概述

2. 机器学习基础

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5. 学习资源和结语



PART 01

人工智能和机器学习概述



什么是机器学习



机器学习致力于如何通过**计算**的手段，利用经验来**改善系统自身的性能**。在计算机系统中，“经验”通常以“数据”形式存在，因此，机器学习所研究的主要内容，是关于在计算机上从数据中产生“模型”（model）的算法，即“学习算法”（learning algorithm）。如果说计算机科学是研究关于“算法”的学问，那么类似的，机器学习是研究关于“**学习算法**”的学问。

摘自西瓜书，周志华教授，南京大学

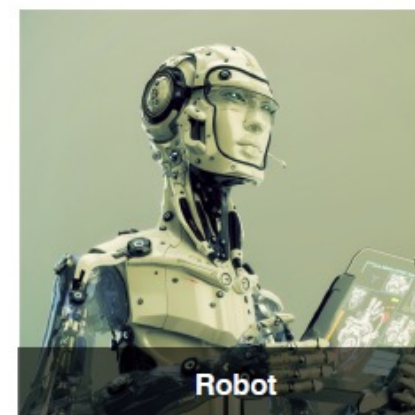
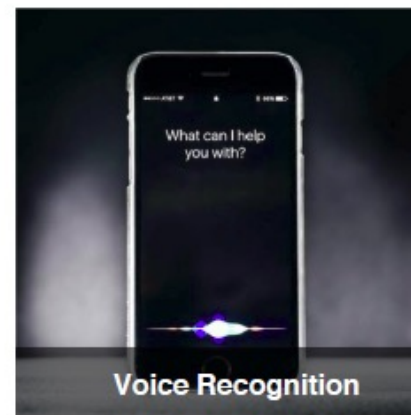
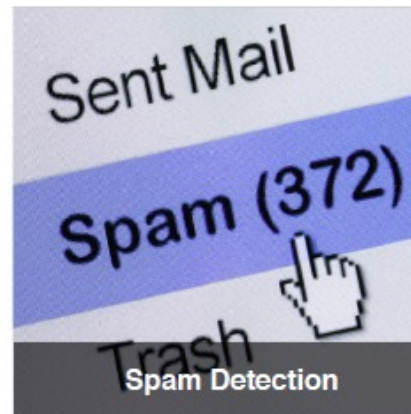
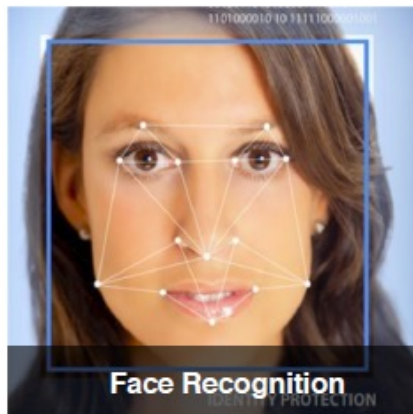
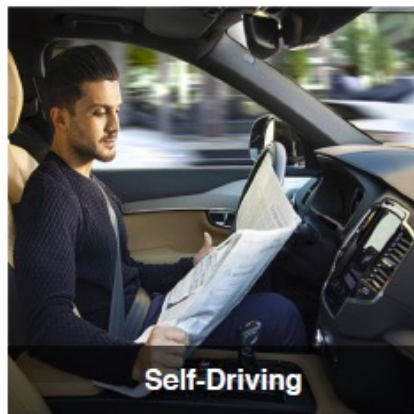
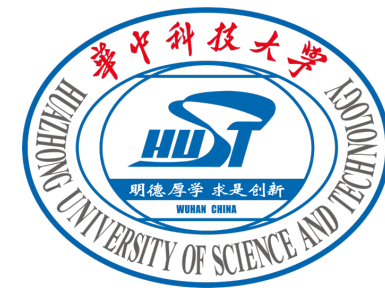
Herbert Alexander Simon (1975年图灵奖，1978年诺贝尔经济学奖)

“Learning is any process by which a system improves performance from experience”

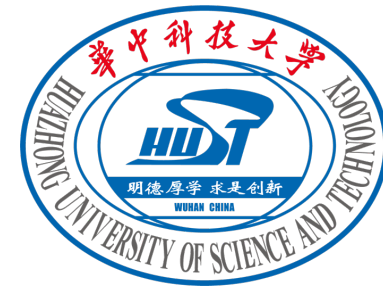




日常生活中的机器学习

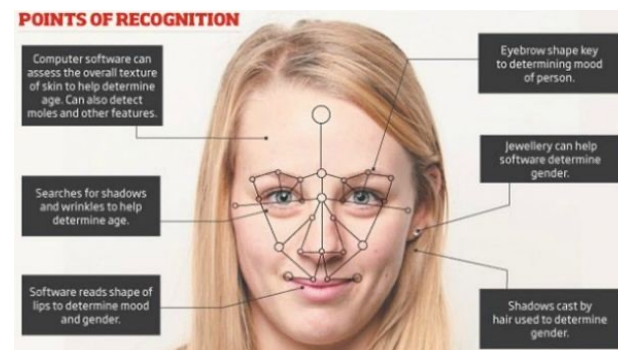


■ 机器学习的应用

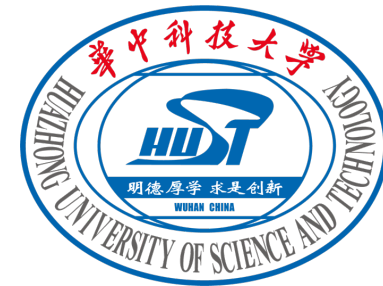


• 计算机视觉 (Computer Vision, CV):

- 获取、处理、分析和理解数字图像和视频
- 应用实例
 - 面部识别: Face ID(苹果)
 - 光学字符识别(Optical Character Recognition, OCR): 数据录入、自动车牌识别等
 - 图片搜索(谷歌)
 - 检测事件:视觉监视或人的计数等
- 用于组成更负责的智能系统
 - 无人驾驶汽车
 - 机器人
- 模仿人类的视觉系统
- 相当发达的领域, **超人**的表现!



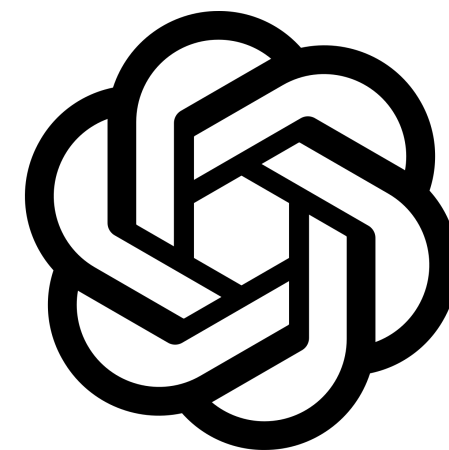
■ 机器学习的应用



• 自然语言处理 (Natural Language Processing, NLP):

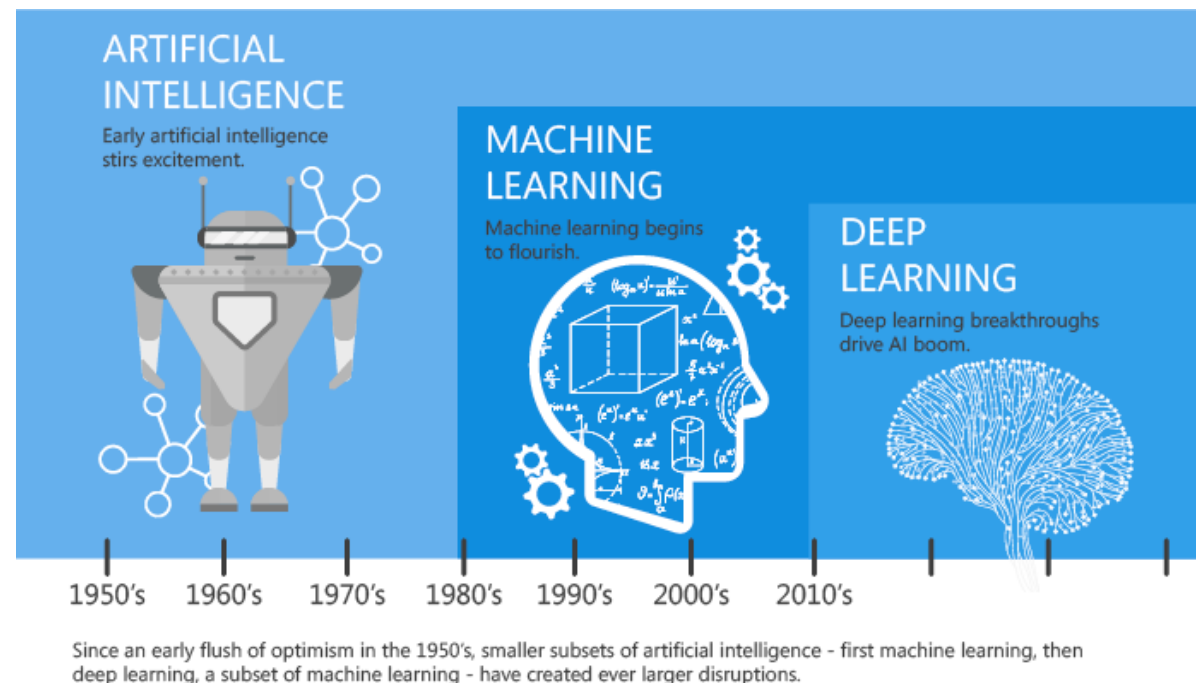
- 计算机与人类自然语言之间的交互
- 应用实例：
 - 智能音箱: 亚马逊Echo、谷歌Home和苹果HomePod
 - 智能手机: Siri(苹果)和Cortana(微软Windows OS)
 - 自动机器翻译
 - 问答系统
 - ChatGPT、Deepseek

- 模仿人类的阅读系统
- 近年来，取得了显著的进展



■ 机器学习和人工智能的历史

- 1950: Alan Turing “Computing Machinery and Intelligence” 和图灵测试
- 1956年诞生于达特茅斯会议 (Dartmouth College)
- 1970-1980: 人工智能寒冬 (AI winter)
- 1980s: 专家系统 (if-then rules)
- 1990s-21世纪: 伴随计算硬件和算法的飞速发展而**快速增长**
 - 1997年IBM研发Deep Blue在**国际象棋**中击败人类世界冠军
- 2010至今: **深度学习**的时代
 - 2017年, DeepMind研发AlphaGo在**围棋**中击败人类世界冠军
 - 2022年底, OpenAI发布ChatGPT



■ 1956年AI诞生于达特茅斯会议 Dartmouth College summer AI conference

1. Automatic Computers
2. How Can a Computer be Programmed to Use a Language
 - It may be speculated that a large part of human thought consists of manipulating words according to rules of reasoning and rules of conjecture. From this point of view, forming a generalization consists of admitting a new word and some rules whereby sentences containing it imply and are implied by others. This idea has never been very precisely formulated nor have examples been worked out.
3. Neuron Nets (theory and practice)
4. Theory of the Size of a Calculation (complexity analysis)
5. Self-improvement
6. Abstractions
7. Randomness and Creativity: the difference between creative thinking and unimaginative competent thinking lies in the injection of some randomness?

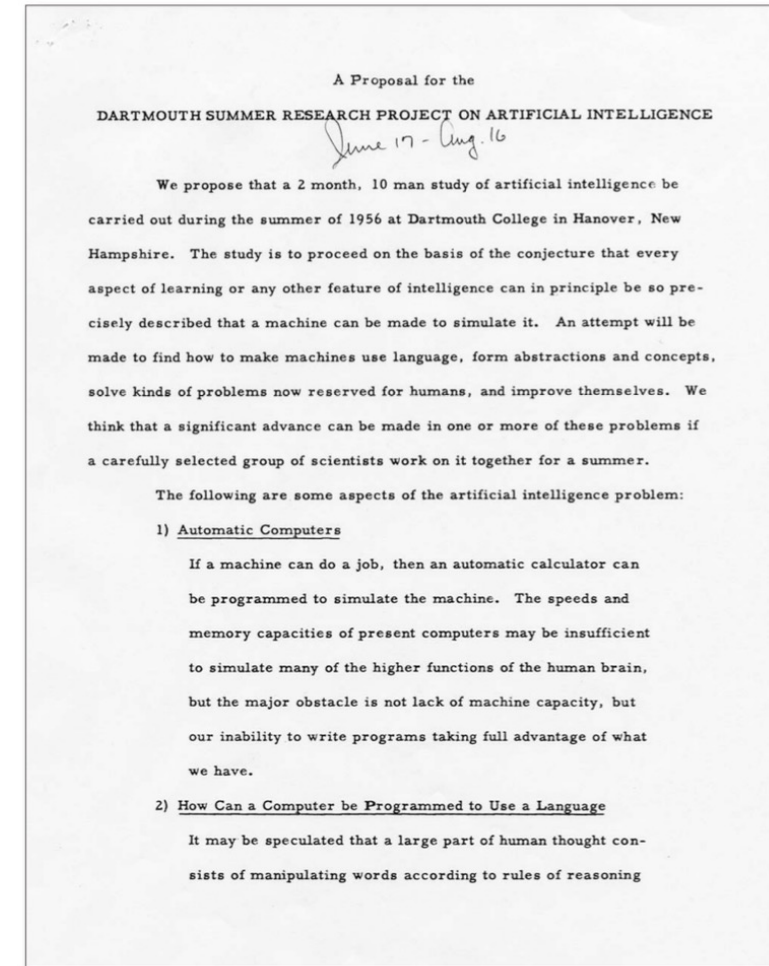
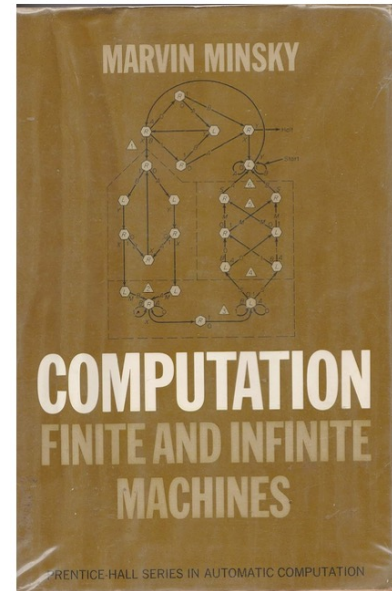
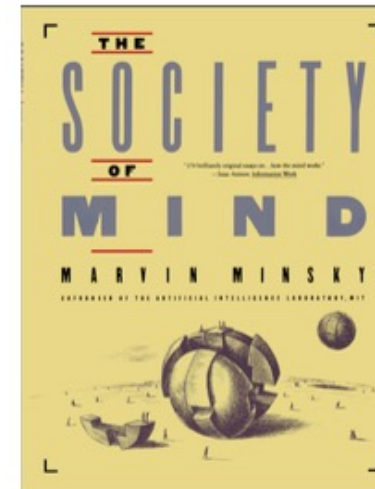
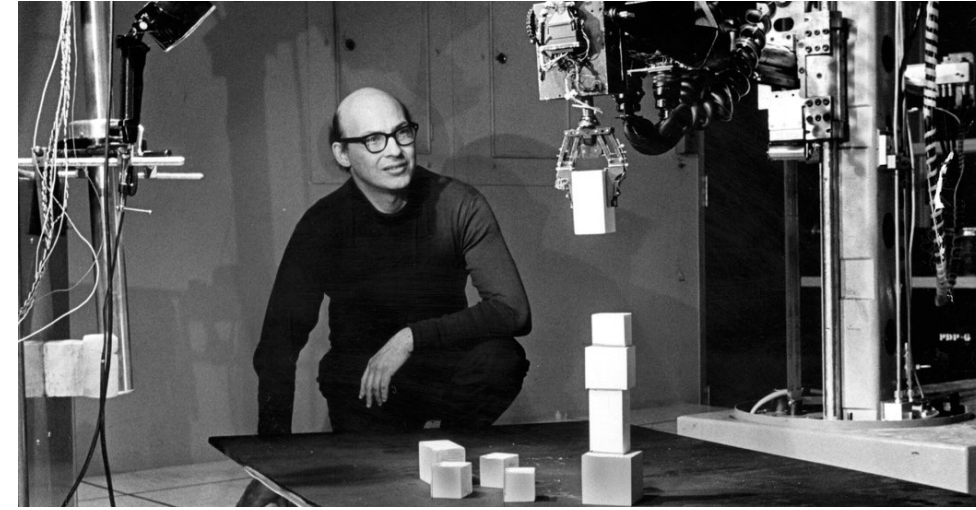


Photo courtesy Dartmouth College.

■ 人工智能的先驱 Marvin Minsky

- 1969: Marvin Minsky **“Computation: Finite and Infinite Machines”**
 - understand the capabilities and their limitations of computers: what all these machines can do and what they cannot do
- 1987: Marvin Minsky **“The Society of Mind”**
 - what magical trick makes us intelligent? The trick is that there is no trick. The power of intelligence stems from our vast diversity, not from any single, perfect principle. (mind as a society of **agents** 智能体)



■ 2006: The Dartmouth Artificial Intelligence Conference: The Next 50 Years AI@50

- Marvin Minsky: "AI can never be a science until it publishes what fails as well as what succeeds."
- In some respects the 1956 summer research project fell short of expectations. The participants came at various times and worked on their own projects, and hence it was not really a conference in the usual sense. There was no agreement on a general theory of the field and in particular on a general theory of learning.
- **The field of AI was launched not by agreement on methodology or choice of problems or general theory, but by the shared vision that computers can be made to perform intelligent tasks.**
- **Evaluations at 2006**: Although AI has enjoyed much success over the last 50 years, numerous dramatic disagreements remain within the field. Different research areas frequently do not collaborate, researchers utilize different methodologies, and there still is no general theory of intelligence or learning that unites the discipline.
- One of the disagreements that was debated at AI@50 is whether AI should be logic based or probability based.
 - Eugene Charniak: "Statistics has taken over natural language processing because it works."



■ 苦涩的一课 The Bitter Lesson by Rich Sutton (2019)

计算机科学和人工智能发展过程中一个痛苦但重要的教训：**随着计算能力的增长，基于通用计算的方法最终总会胜过依赖人类知识的“人工”设计方法**

1. 经验教训：人类手工设计 vs. 计算能力

- 在过去 70 年的计算机科学发展中，研究人员反复尝试通过人类智慧手工设计特定领域的规则和知识，以提升智能系统的表现。
- 然而，**历史反复证明**，这些手工设计的方法最终都被基于计算能力的通用方法超越。
- 例如，在计算机视觉、语音识别、象棋游戏等领域，手工特征工程和专家系统最终被**数据驱动**的方法取代

2. 计算能力的主导地位

- 最成功的方法都是那些**可以利用大量计算资源**的方法，而不是依赖人类先验知识的方法
- 例如，深度学习（Deep Learning）并没有在特定领域注入大量专家知识，而是**利用数据和计算能力自动学习**。
- 近年来，AlphaGo、GPT 等人工智能系统的成功，正是基于大规模计算和数据驱动的方法，而非专家知识。

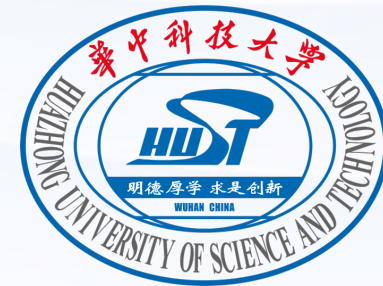
3. 结论和未来的发展方向

- **计算能力的增长**仍是人工智能发展的核心驱动力。
- **通用学习方法（如深度学习）**比手工设计的专家知识更有效，未来的AI系统应该尽可能少地依赖人类知识，而更多依赖计算和数据驱动的方法。
- 对于新问题，最好的策略是找到一个**可扩展**的学习方法，并**利用最大可能的计算资源**
- 未来的突破不会来自特定领域的专家，而会来自于**利用计算能力和数据的通用方法**。



强化学习之父

<http://www.incompleteideas.net/IncIdeas/BitterLesson.html>

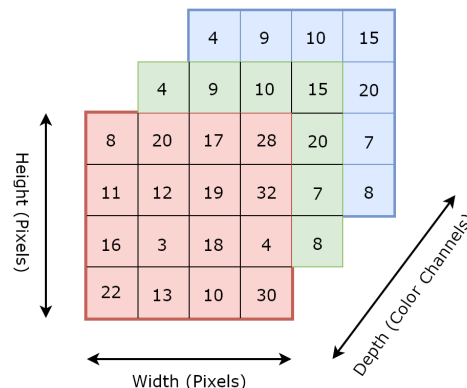


PART 02

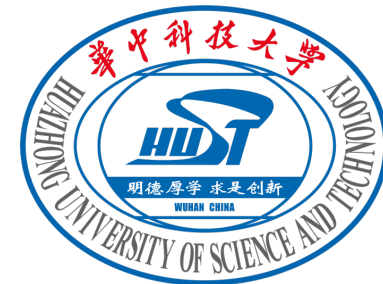
机器学习基础

为什么不是专家系统

- **专家系统**: 由人类专家制定规则, 电脑执行if-else判断
 - 从计算角度, **不可能**完成围棋之类的**复杂**任务
 - 为什么?
 - 围棋棋盘被划分成19 x 19的网格
 - 搜索和决策空间19 x 19 = 361 (针对**每一步**!)
 - 如果仅考虑15步, 则有 $361^{15} = 5.3e76$ 种不同的的走法
 - 现代物理: 宇宙中有大约 $1e80$ 原子
 - 当搜索空间是**连续**时:
 - 自动驾驶: 可以选择在**任何**时间将方向盘转动**任何**角度
- 当 (人为) 寻找有效规则非常困难时
 - 猫和狗的图像、照片分类



■ 专家系统举例



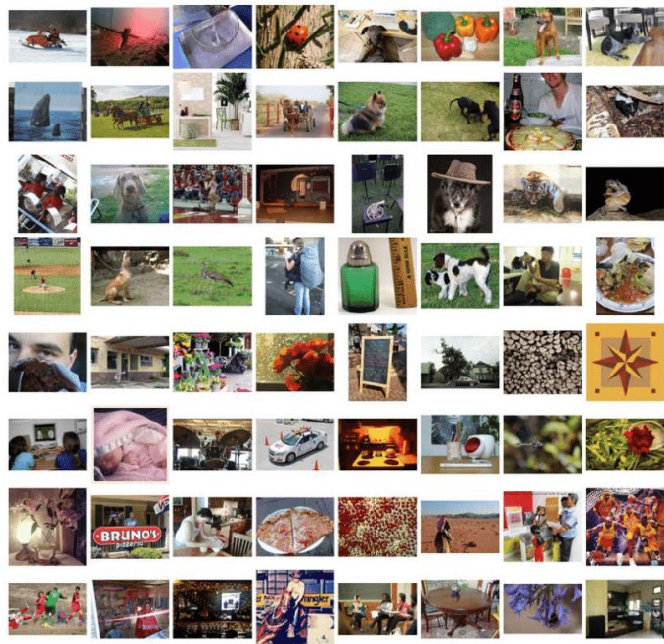
- 举例：房价预测问题

MSSubC lass	MSZoning	LotFront age	LotArea	Street	Alley	LotShape	LandCon tour	Utilities	HouseStyle	Price
20	RH	80	11622	Pave	NA	Reg	Lv1	AllPub	1 Story	65k

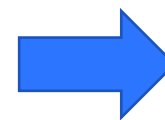
- 专家系统：
 - $price = 0$
 - **If** HouseStyle=1 Story **and** LotArea $\leq$ 8000, **then** price=LotArea*5
 - **If** HouseStyle=1 Story **and** LotArea $>$ 8000, **then** price=LotArea*6
 - **If** HouseStyle=2 Story **and** LotArea $\leq$ 12000, **then** price=LotArea*6.5
 - ...
 - **if** LandContour=Lv1 **and** Street=Pave, **then** price=price-6k
- 数据来自<https://www.kaggle.com/alphaepsilon/housing-prices-dataset>.

■ 数据驱动 (data driven) 的机器学习

- 从大量数据中**自我学习**、总结特征（而不是依靠人类专家设计）
- **数据量较大时**，**有可能**找到比人类专家设计的更好的特征



ML

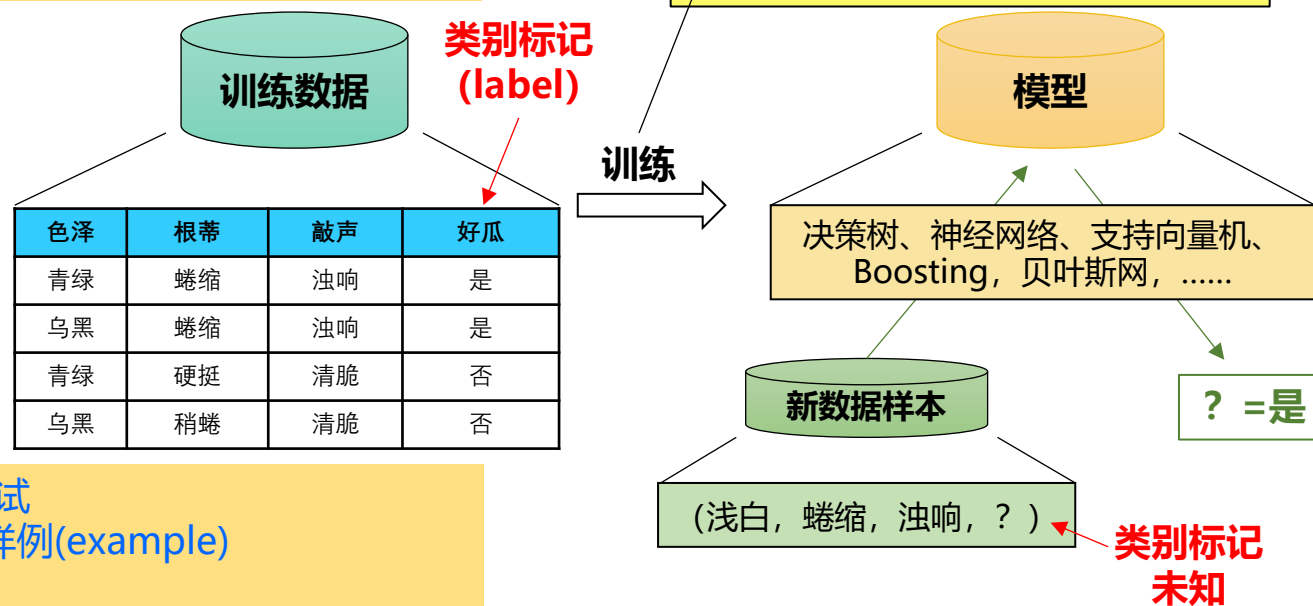


Labels:

- dogs
- cats
- trains
- aircrafts
- etc.

机器学习的分类和基本术语

- 监督学习(supervised learning)
- 无监督学习(unsupervised learning)
- 半监督学习(semi-supervised learning)



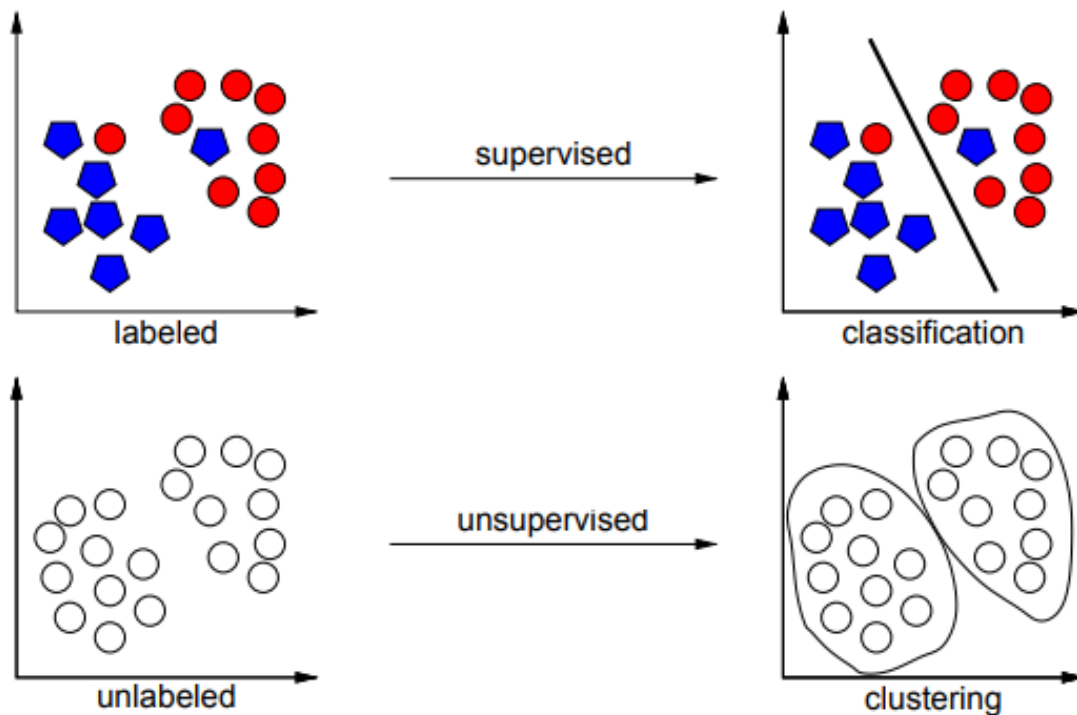
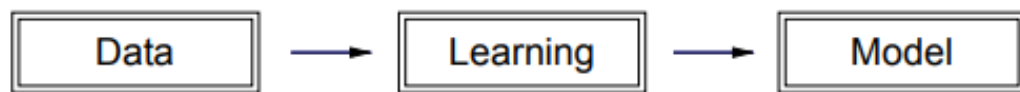
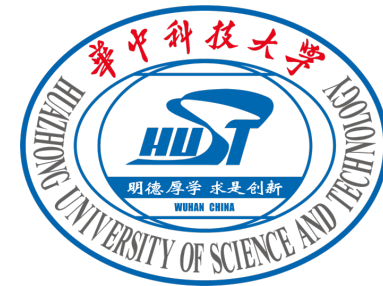
- 数据集: 训练, 测试
- 示例(instance), 样例(example)
- 样本(sample)
- 属性(attribute), 特征(feature); 属性值
- 属性空间, 样本空间, 输入空间
- 特征向量(feature **vector**)
- 标记空间, 输出空间

- 未见样本(unseen instance)
- 未知“分布”

例子来自西瓜书, 周志华教授, 南京大学



机器学习的分类和基本术语



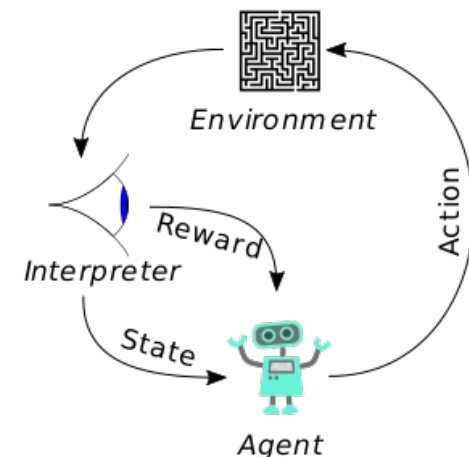
监督学习

半监督学习

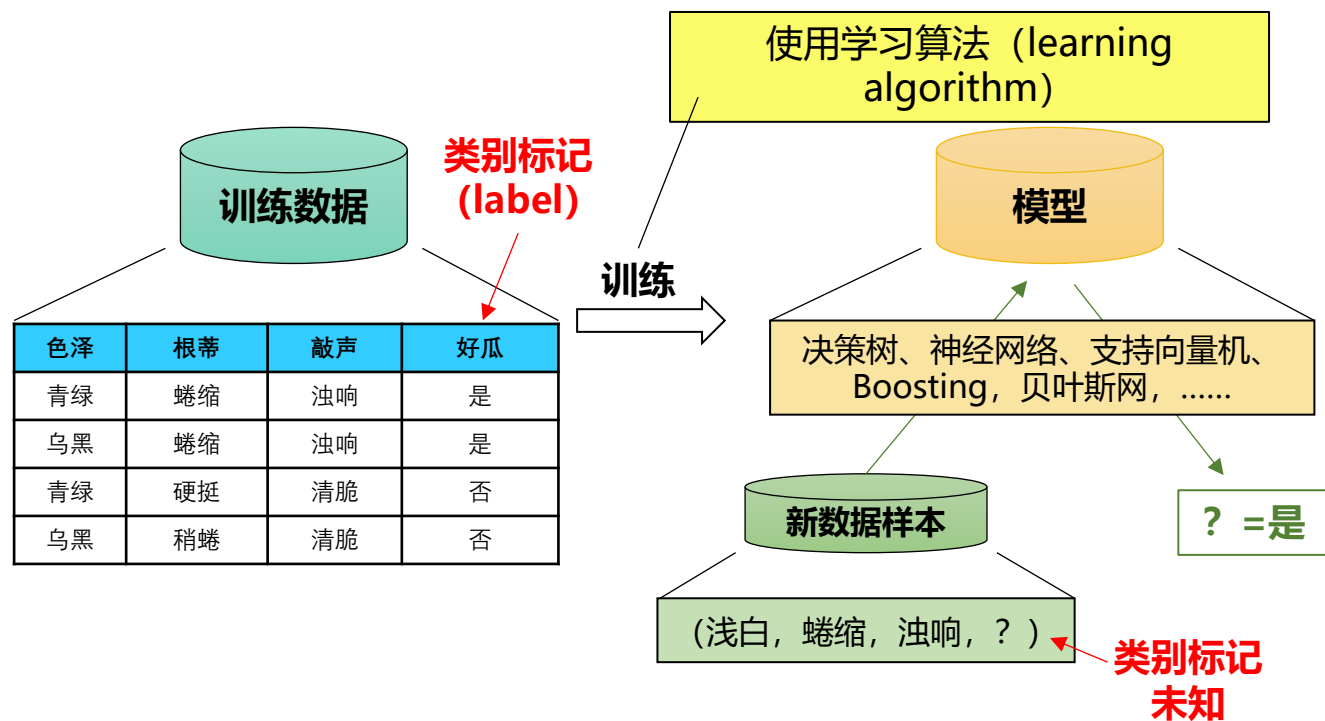
无监督学习

强化学习：从“尝试—反馈”（而非标签）中学习：

https://www.bilibili.com/video/BV1tm4y1h79M/?share_source=copy_web&vd_source=405ba20a8a5d0e3e737677cdbaa02da6



■ 典型的机器学习过程



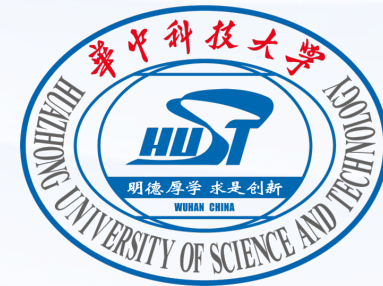
什么模型好?

泛化能力要强

能很好的适用于unseen instance, 如, 错误率低

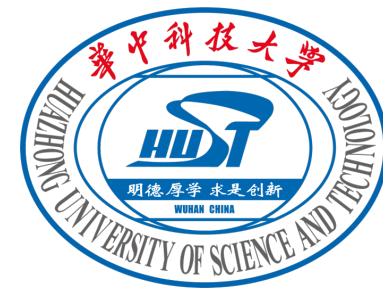
然而, 我们手上没有unseen instance

模型训练—模型评估—模型推理



PART 03

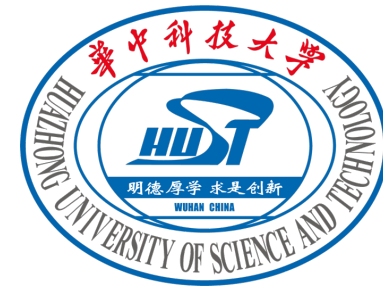
从线性模型到神经网络



■ 机器学习的本质

- Machine Learning =
Representation + Evaluation + Optimization
 - 来源: Pedro Domingos, Communications of the ACM, "*A Few Useful Things to Know About Machine Learning*"
 - 好的“表示” (representation) 或者特征 (feature) 是人工智能-机器学习的**关键!** (也是和传统专家系统的**本质区别**)
 - 如何获得 **“好的”** representation?
 - 首先需要定义、判断什么是 **“好的”** : evaluation (模型评估)
 - 一旦确定了什么是 **“好的”** , 转化为一个优化 (optimization) 问题求解

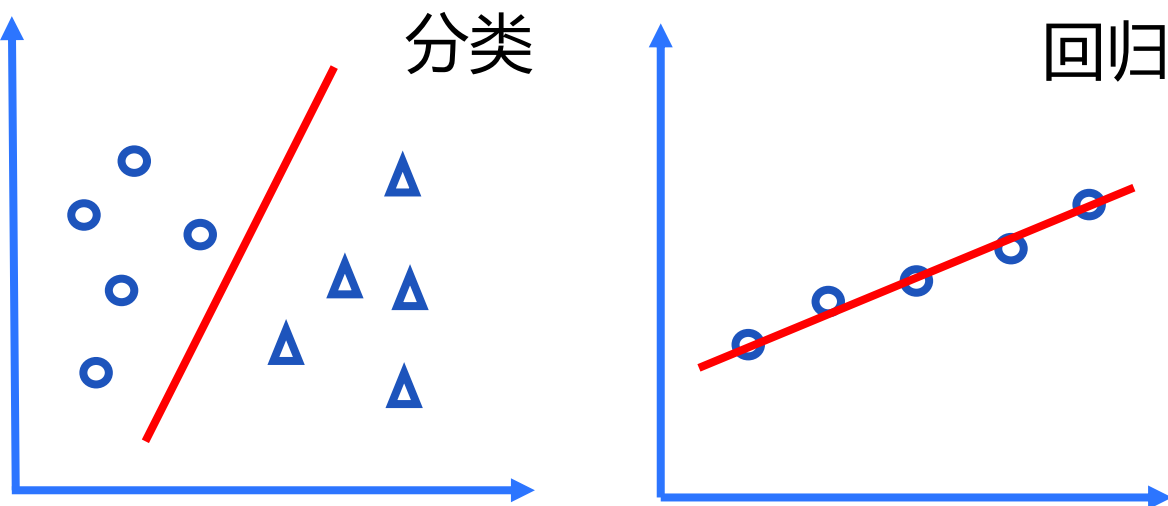
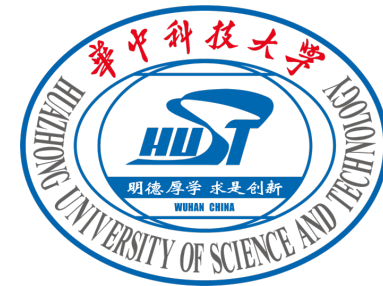
■ 机器学习、人工智能模型训练的流程



1. **数据**：足够多的优质数据是根本保证
2. **模型选择**：把黑盒子的模型输出—输出关系**转化为参数的确定**
3. **模型评估**（怎么定义“好”），设计损失函数
4. **模型训练、优化**



线性模型



向量形式：

$$f(x) = w^T x + b$$

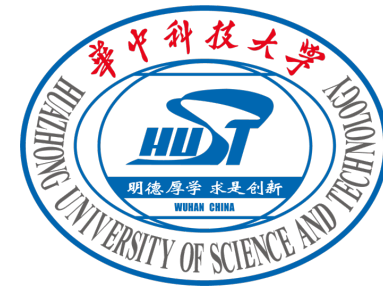
简单，可解释性好

线性模型：从数据中学得一个通过属性（特征）的线性组合来进行预测的函数

$$f(x) = w_1 x_1 + w_2 x_2 + \dots + w_d x_d + b$$

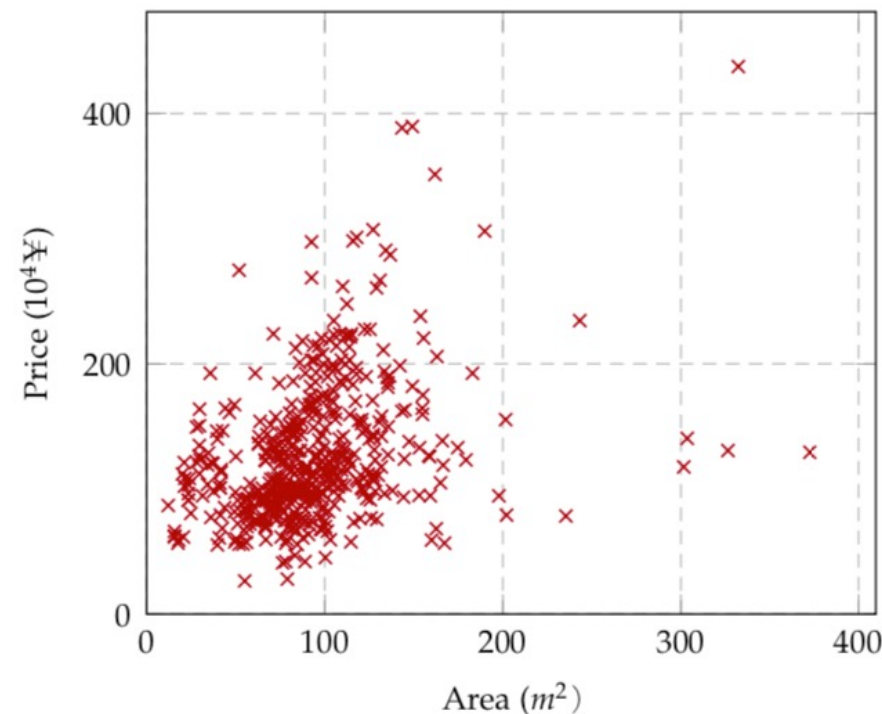


例子：房价预测



• **问题:** 对于一套 180 m^2 的房子，我们应该报价多少钱？

1. 收集历史数据
2. 选择自己的模型
3. 用数据拟合模型（算法）
4. 作出预测



数据来自: <https://www.kaggle.com/alphaepsilon/housing-prices-dataset>



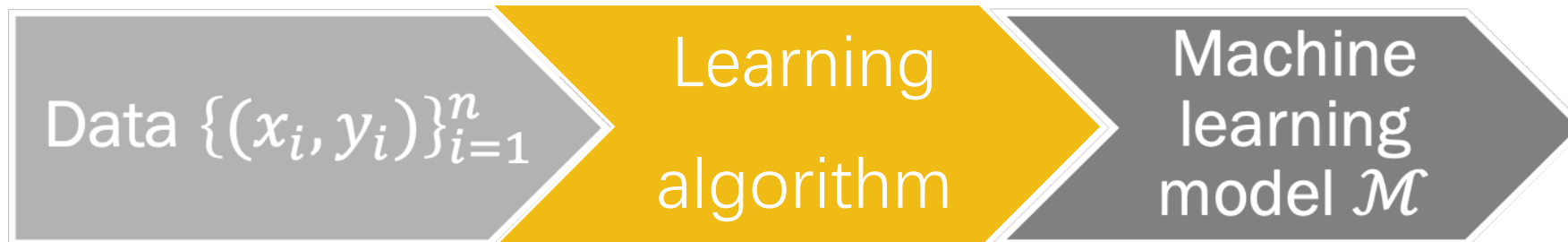
房价预测：数据



No.	面积 (m ²): x	价格 (10 ⁴ RMB): y
1	78.50	145.95
2	88.72	98.00
3	93.68	214.90
...	...	...

- 总共获得 n (历史)数据(面积, 价格)
- x_i : 数据/特征 (输入)
- y_i : 目标 (输出)
- **目的:**
 - 对于新输入 $\hat{x}$, 预测它的 $\hat{y}$

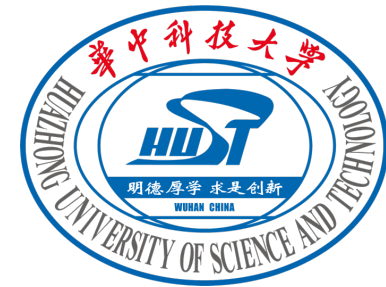
流程图



做出预测: $\hat{x} \rightarrow \mathcal{M} \rightarrow \hat{y}$



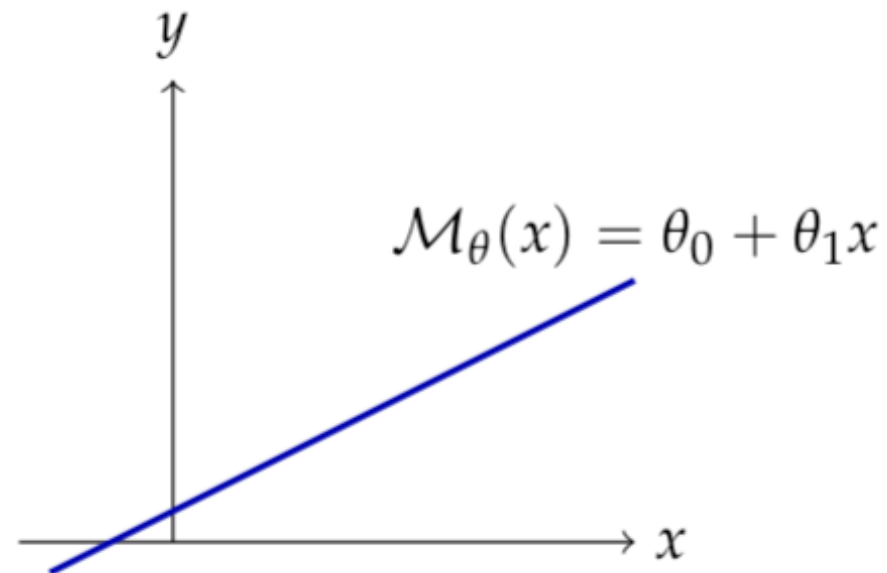
房价预测：模型



$$y: \mathcal{M}_\theta(x) = \theta_0 + \theta_1 x$$

- 描述了输入 x 和输出 y 之间的一个线性关系
- 模型 $\mathcal{M}_\theta$ 由 (θ_0, θ_1) 确定
- 如何决定模型 $\mathcal{M}_\theta$ 是否是好的?

=> 目标函数!

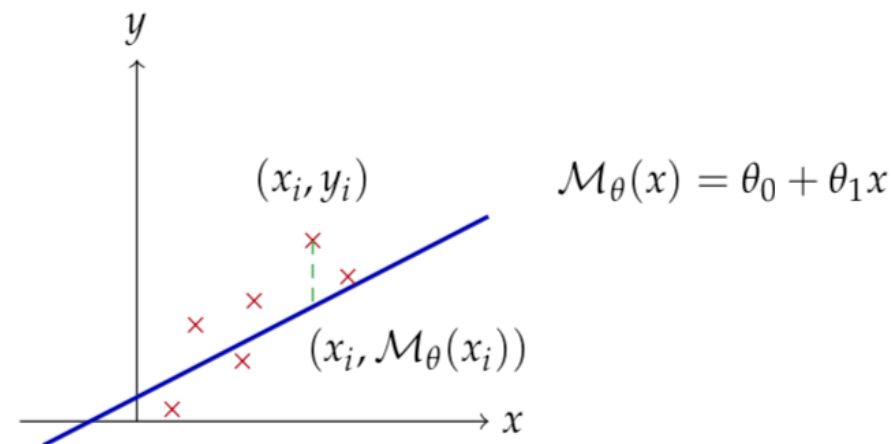


房价预测：模型评估

- 线性回归 $\mathcal{M}_\theta(x) = \theta_0 + \theta_1 x$, 由两个参数 (θ_0, θ_1) 决定
- 对于数据 $\{(x_i, y_i)\}_{i=1}^n$, 目标/损失函数 $L(\theta_0, \theta_1)$ 告诉我们模型（参数）如何**拟合**数据

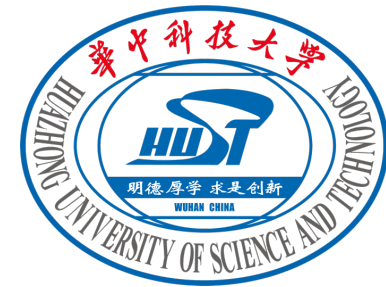
$$\begin{aligned} L(\theta_0, \theta_1) &= \frac{1}{n} \sum_{i=1}^n (y_i - \mathcal{M}_\theta(x_i))^2 \\ &= \frac{1}{n} \sum_{i=1}^n (y_i - \theta_0 - \theta_1 x_i)^2 \end{aligned}$$

- $L(\theta_0, \theta_1) \uparrow$, 模型输出和历史数据之间的**差异更大**
- 找到**最好的** $(\theta_0, \theta_1) \Rightarrow \min_{\theta_0, \theta_1} L(\theta_0, \theta_1)$

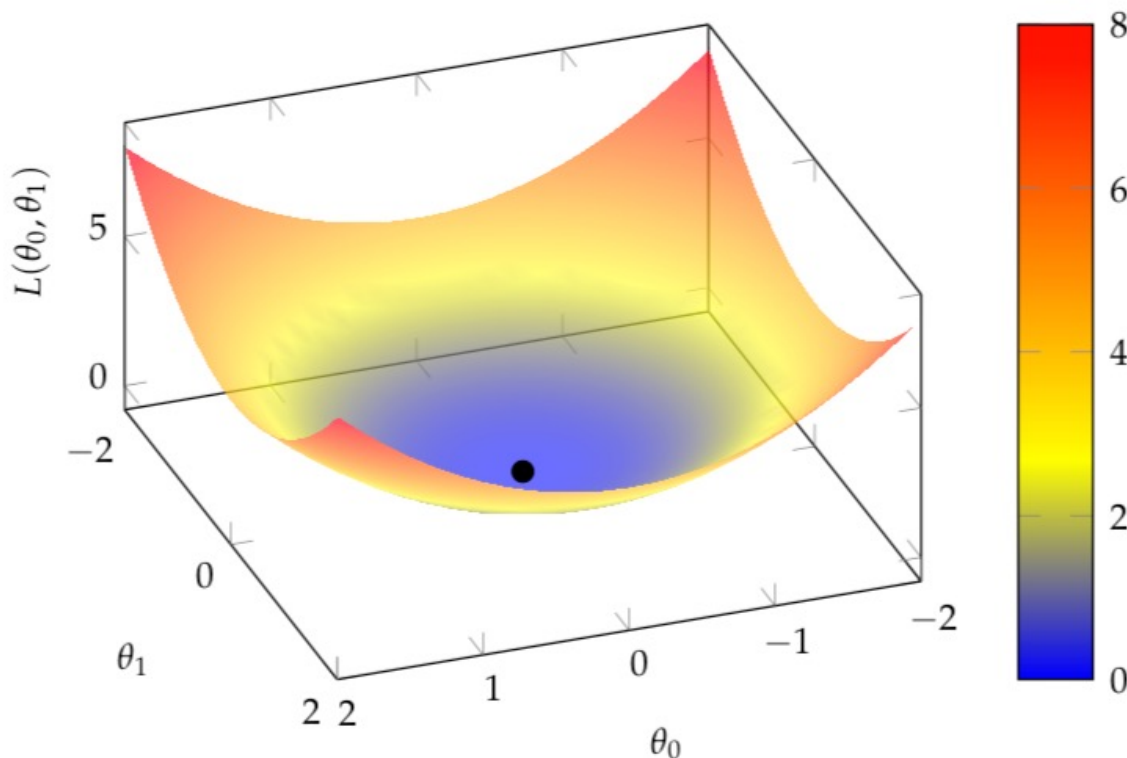




房价预测：优化

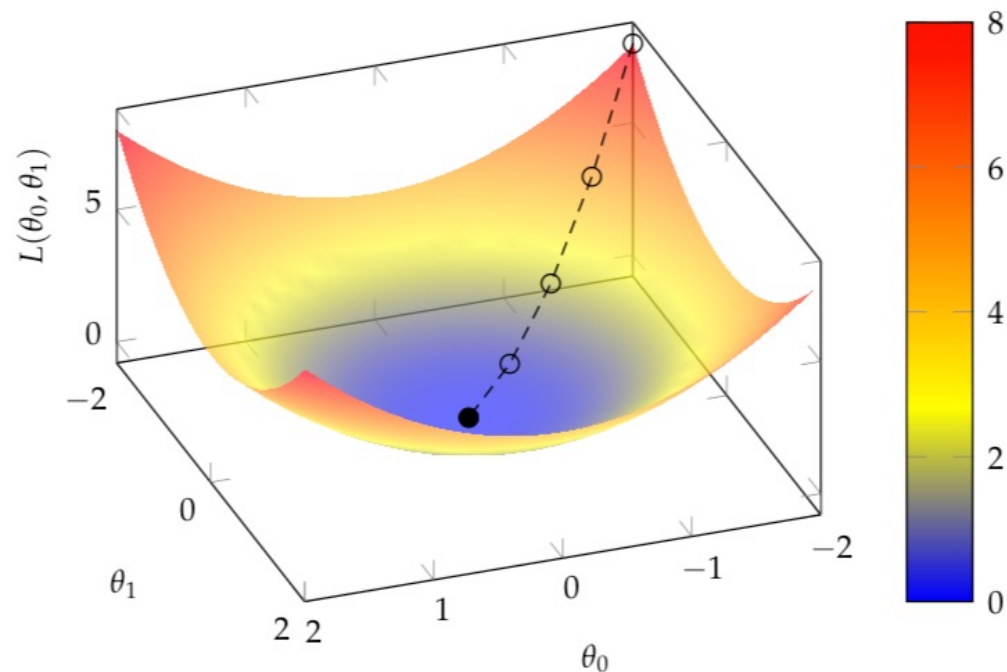
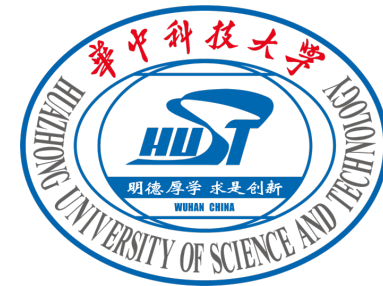


- 对于给定的数据集 $\{(x_i, y_i)\}_{i=1}^n$, 绘制 $L(\theta_0, \theta_1)$ 如下
- 希望到达底部（黑色）点 $\min L(\theta_0, \theta_1)$, 获取相应的 (θ_0^*, θ_1^*) 值





房价预测：优化

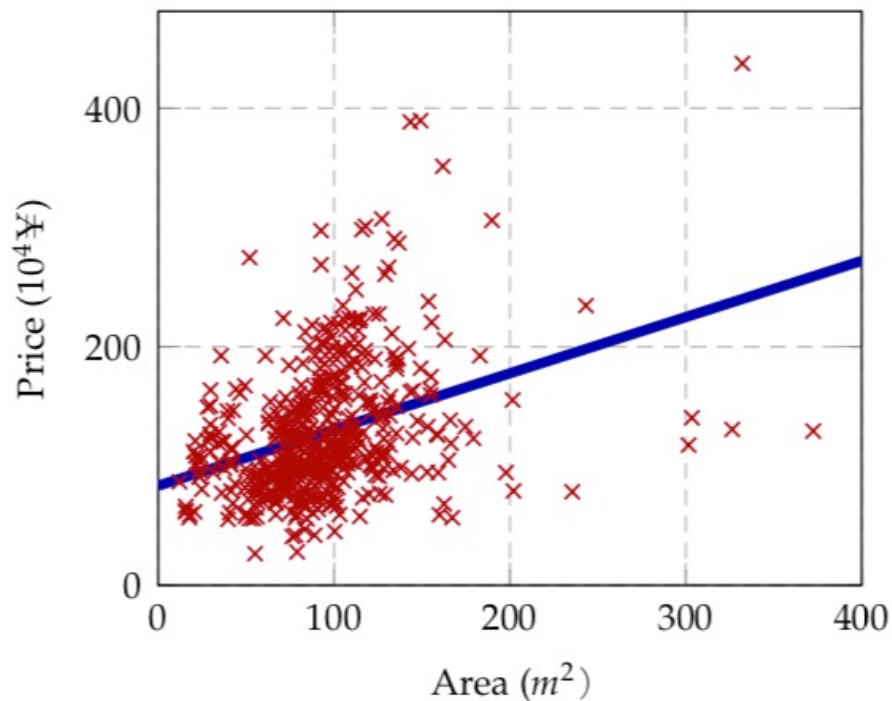
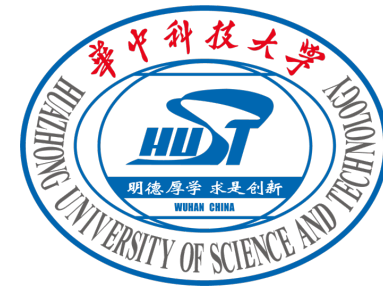


算法

- 从任意点 (θ_0^0, θ_1^0) 开始, **更新** (θ_0^k, θ_1^k) 来减小 $L(\theta_0, \theta_1)$, 以达到最小的 (θ_0^*, θ_1^*) : $L(\theta_0^{k+1}, \theta_1^{k+1}) \leq L(\theta_0^k, \theta_1^k)$.
- 根据 $\theta_j^{k+1} = \theta_j^k - \alpha \nabla_{\theta_j} L(\theta_0, \theta_1), j = 0, 1$ 进行迭代更新, 直到收敛



房价预测：预测

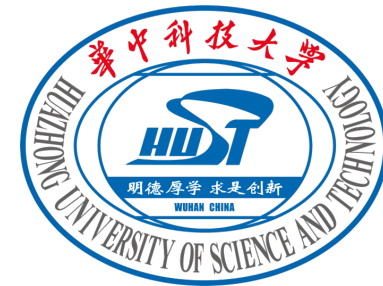


$$\Rightarrow \theta_0^* = 83.5384, \theta_1^* = 0.4734.$$

对于一套 180 m^2 的房子，我们的模型预测 $\mathcal{M}_{\theta^*}(180) = \theta_0^* + 180 \theta_1^* =$
 $168.75 \times 10^4 \text{ RMB}$



房价预测例子：总结



该例中：

1. 线性回归模型：输入输出**线性**关系
2. 均方误差：关于**回归**问题
3. 梯度下降优化方法

当然我们可以有更多选择：

1. 对于**非线性**输入-输出关系：多项式回归、SVM、神经网络
2. 对于**分类**问题：0-1误差、交叉熵误差、合页损失等
3. 越来越先进的优化方法

从线性回归到对数线性回归

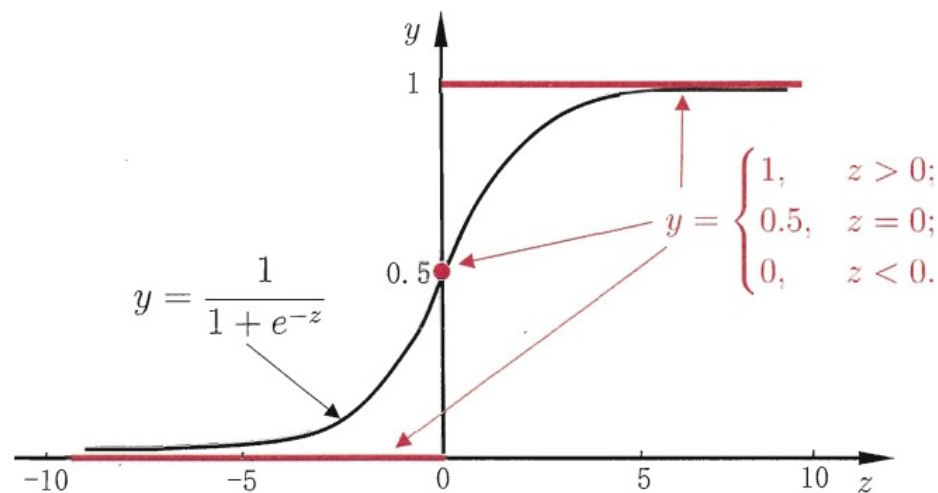
- 线性回归 $y = w^T x + b$ ，刻画了输入 x 和输出 y 之间的线性关系
- 如何描述输入输出之间的非线性关系？比如： $\log(y) = w^T x + b$
- 可推广到一类“广义线性模型”： $g(y) = w^T x + b$ ，选择不同的非线性函数（link function） g

期待输出 $y \in \{0, 1\}$

$$z = w^T x + b$$

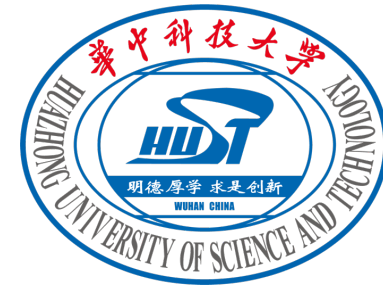
$$y \approx g^{-1}(z) = \begin{cases} 0, & z < 0; \\ 1/2, & z = 0; \\ 1, & z > 0. \end{cases}$$

$$y \approx g^{-1}(z) = \frac{1}{1 + e^{-z}}$$





对数几率回归



$$\log \boxed{\frac{y}{1-y}} = \mathbf{w}^T \mathbf{x} + b$$

损失函数：（来自于最大似然maximum likelihood）
定义

$$\theta = [\mathbf{w}, b], \quad g^{-1}(z) = \frac{1}{1+e^{-z}}, \quad h_{\theta}(x) = \frac{1}{1+e^{-\theta^T x}}$$

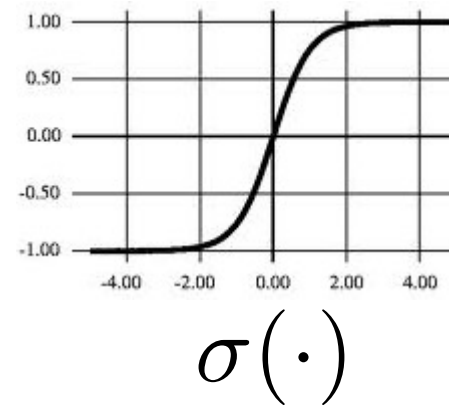
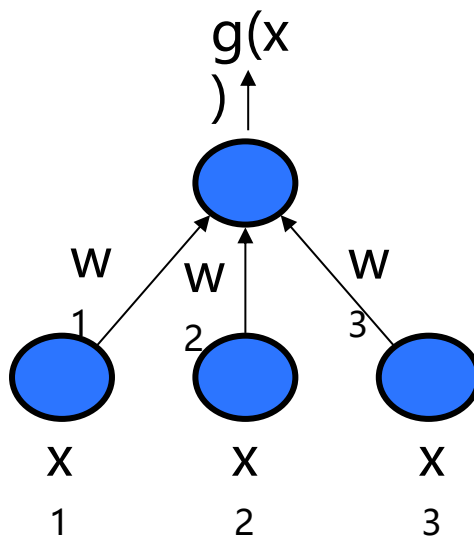
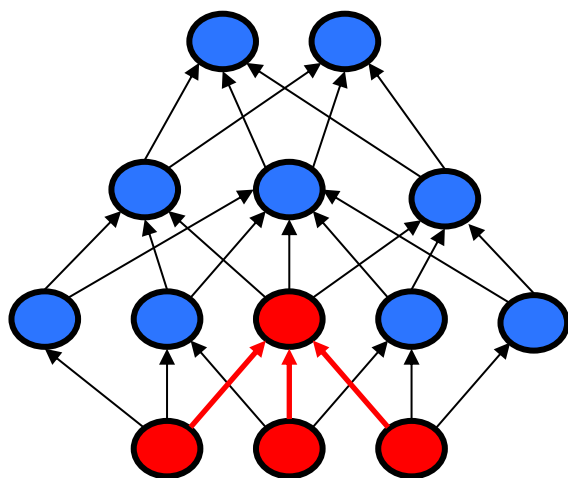
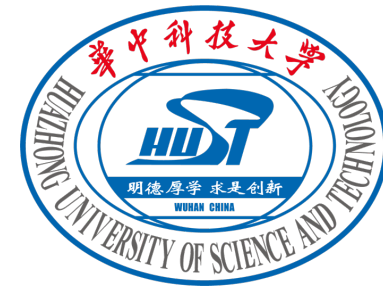
选取 $g(y) = \log \frac{y}{1-y}$
几率 (odds) , 反映
样本作为正例的可能性

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)})$$

$$= -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$



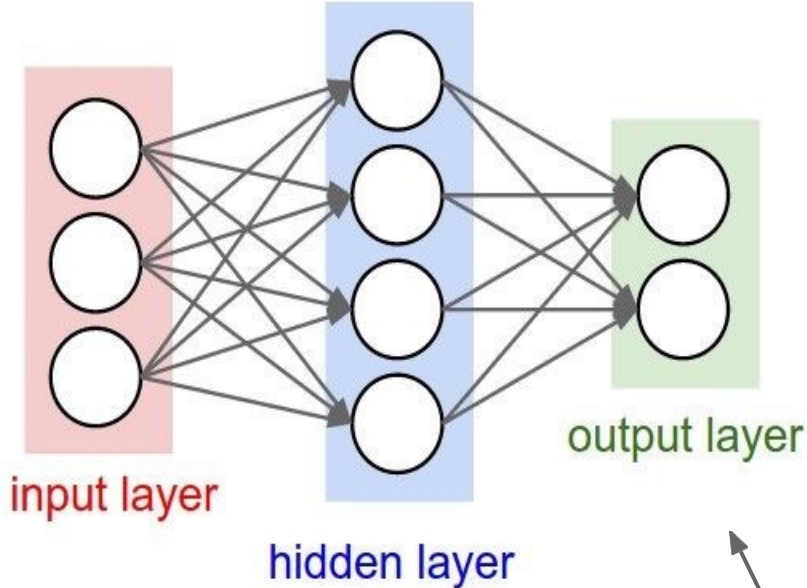
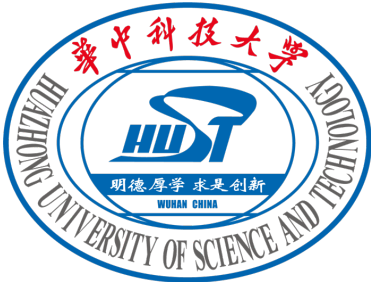
神经网络就是多个对数（或非对数）几率回归



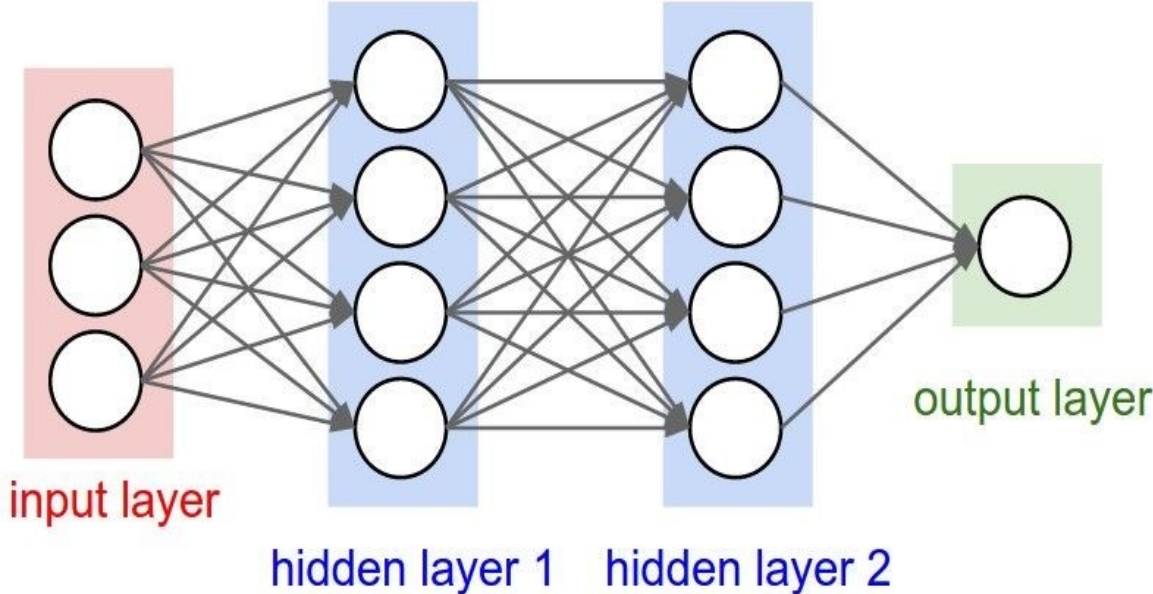
$$g(\mathbf{x}) = \sigma \left(\sum_{i=1}^d x_i w_i + b \right) = \sigma(\tilde{\mathbf{w}}^T \tilde{\mathbf{x}})$$



神经网络结构



"2-layer Neural Net" , or
"1-hidden-layer Net" Neural

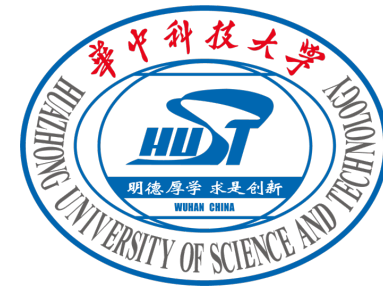


"3-layer Neural Net" , or
"2-hidden-layer Neural Net"

全连接层（和卷积层相对）



反向传播



前向传播

$$z_1 = x^T W_1 + b_1$$

$$a_1 = \tanh(z_1)$$

$$z_2 = a_1^T W_2 + b_2$$

$$\hat{y} = a_2 = \text{softmax}(z_2)$$

$$\frac{\partial \tanh(x)}{\partial x} = 1 - \tanh^2(x)$$

反向传播

$$\frac{\partial L}{\partial W_2} = \frac{\partial z_2}{\partial W_2} \frac{\partial L}{\partial z_2} = a_1^T (\hat{y} - y)$$

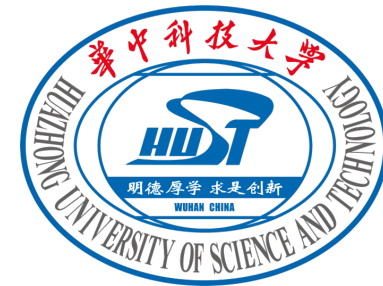
$$\frac{\partial L}{\partial b_2} = \frac{\partial L}{\partial z_2} \frac{\partial z_2}{\partial b_2} = (\hat{y} - y)$$

$$\frac{\partial L}{\partial W_1} = \frac{\partial L}{\partial z_2} \frac{\partial z_2}{\partial a_1} \frac{\partial a_1}{\partial z_1} \frac{\partial z_1}{\partial W_1}$$

$(\hat{y} - y) \quad W_2 \quad (1 - \tanh^2 z_1) \quad x^T$

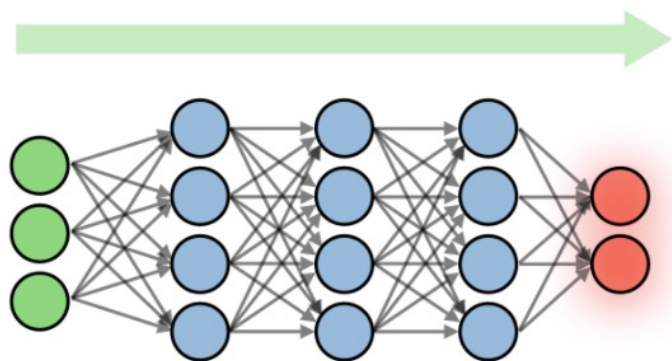


反向传播

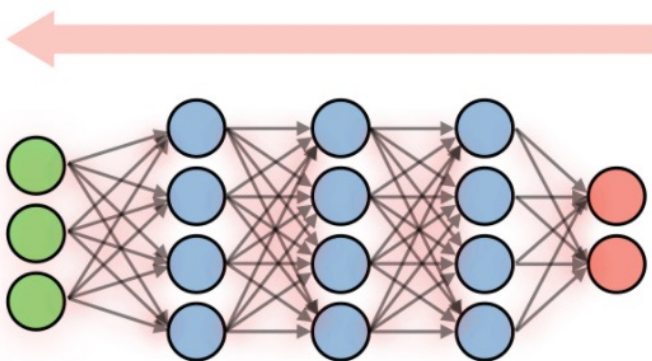


□ **Updating weights** — In a neural network, weights are updated as follows:

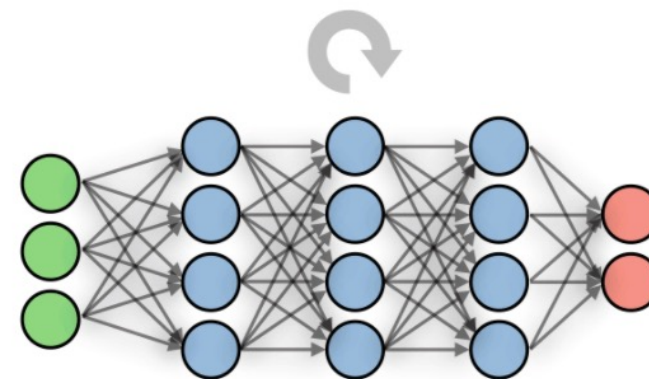
- Step 1: Take a batch of training data and perform forward propagation to compute the loss.
- Step 2: Backpropagate the loss to get the gradient of the loss with respect to each weight.
- Step 3: Use the gradients to update the weights of the network.



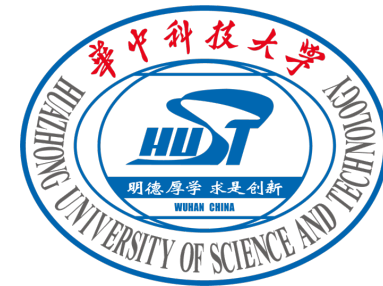
① Forward propagation



② Backpropagation



③ Weights update



PART 04

深度学习与 人工智能的未来

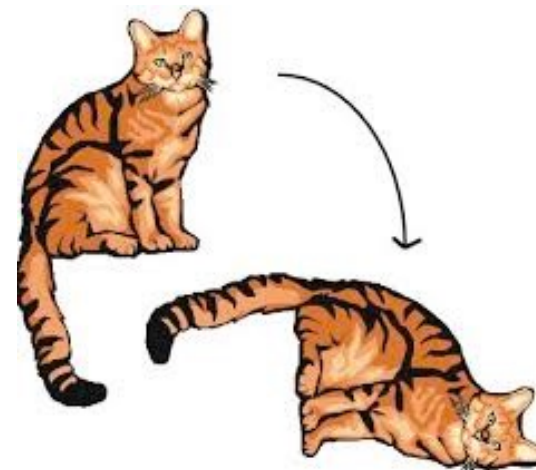
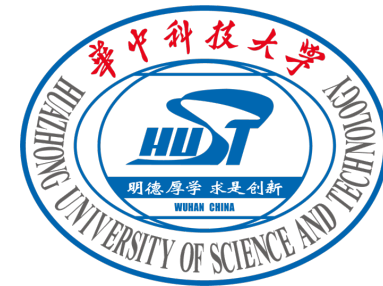
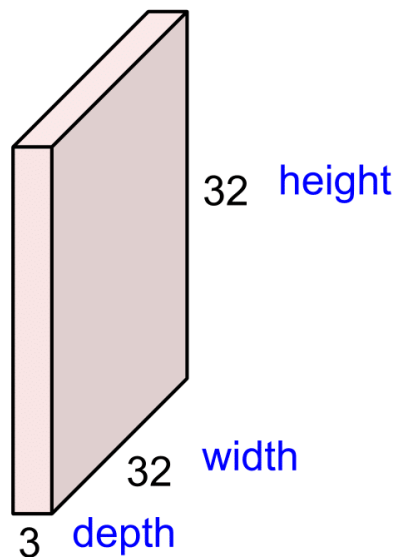


深度学习的结构突破

- 卷积神经网络 (CNN)
- 循环神经网络 (RNN)
- Transformer模型

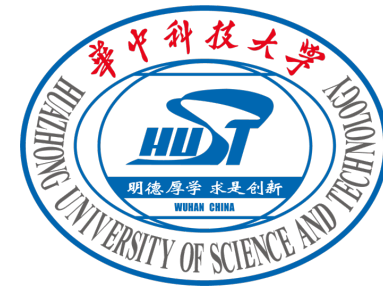
Convolution Layer

32x32x3 image





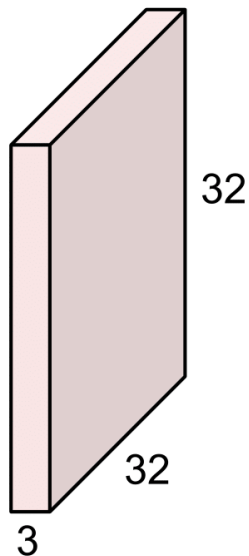
深度学习的结构突破



- 卷积神经网络 (CNN)
- 循环神经网络 (RNN)
- Transformer模型

Convolution Layer

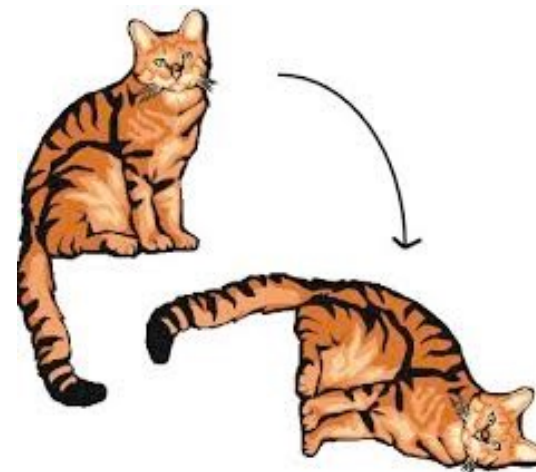
32x32x3 image



5x5x3 filter

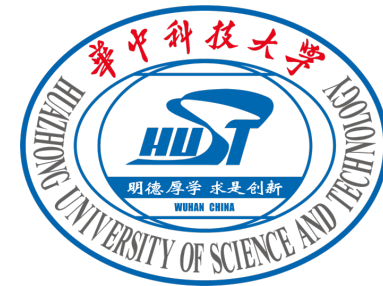


Convolve the filter with the image
i.e. “slide over the image spatially,
computing dot products”





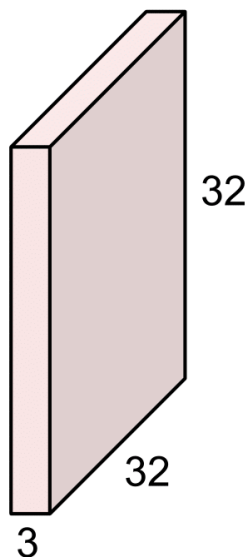
深度学习的结构突破



- 卷积神经网络 (CNN)
- 循环神经网络 (RNN)
- Transformer模型

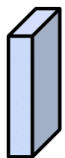
Convolution Layer

32x32x3 image

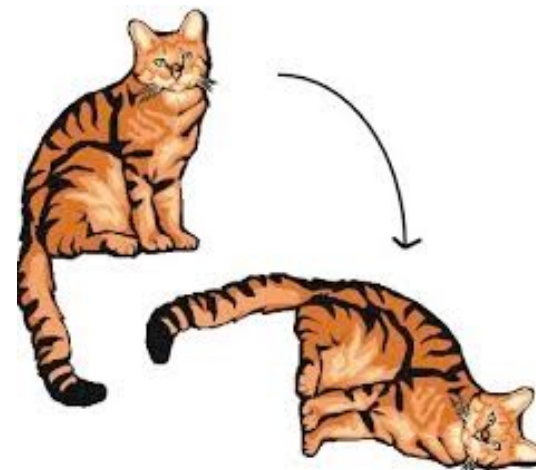


Filters always extend the full depth of the input volume

5x5x3 filter

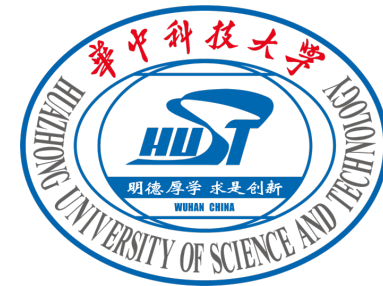


Convolve the filter with the image
i.e. “slide over the image spatially,
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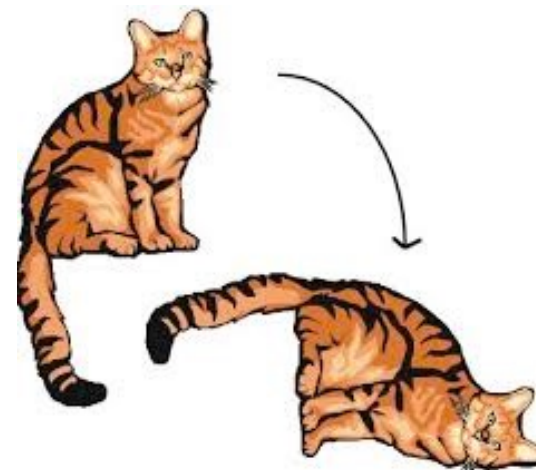
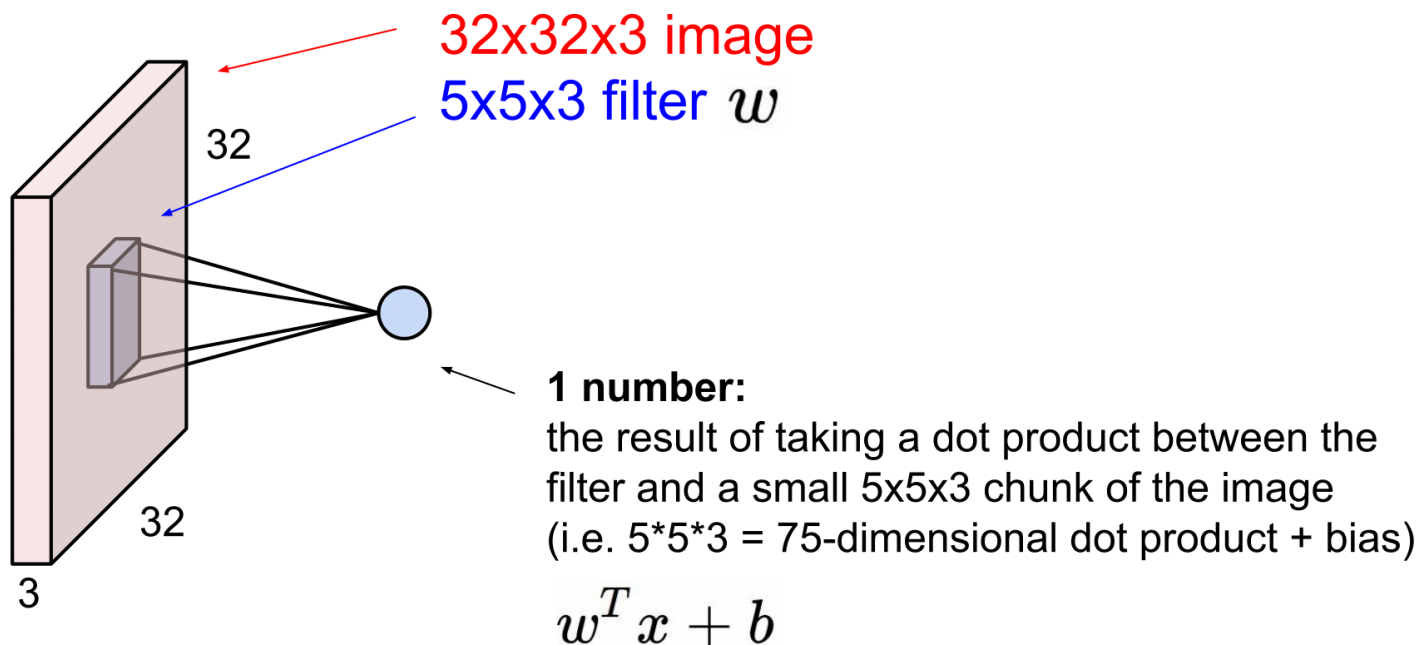


深度学习的结构突破



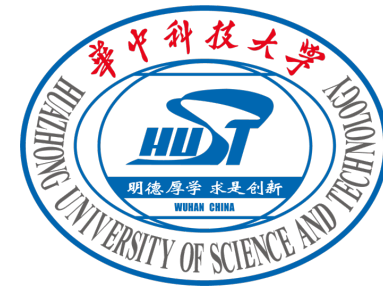
- 卷积神经网络 (CNN)
- 循环神经网络 (RNN)
- Transformer模型

Convolution Layer



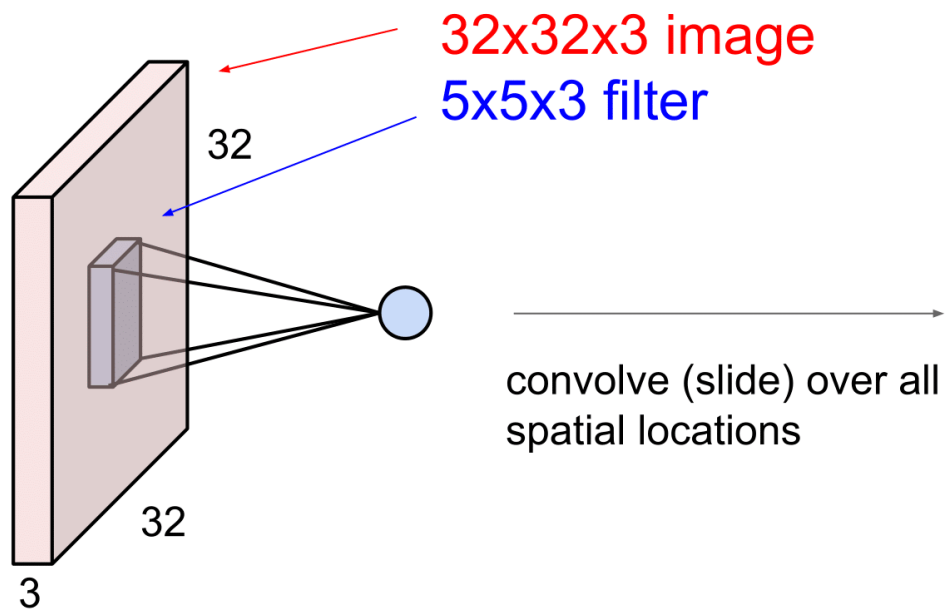


深度学习的结构突破

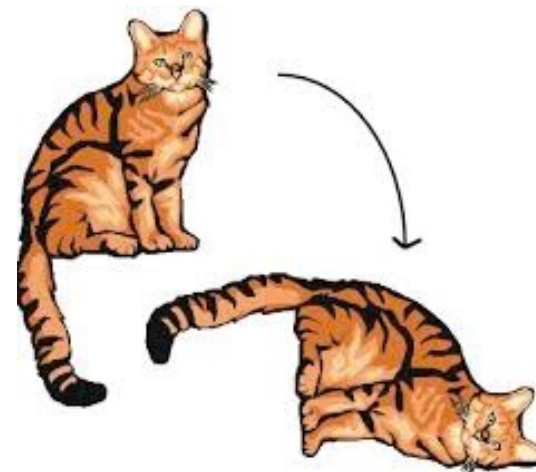
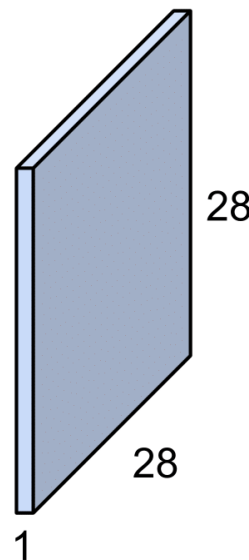


- 卷积神经网络 (CNN)
- 循环神经网络 (RNN)
- Transformer模型

Convolution Layer

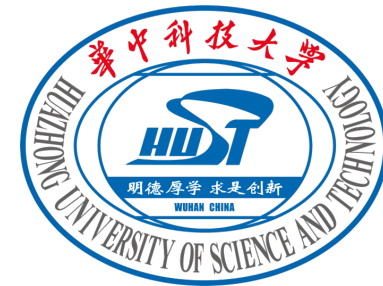


activation map





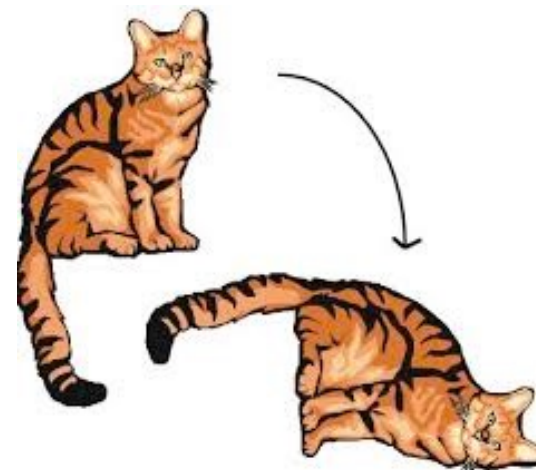
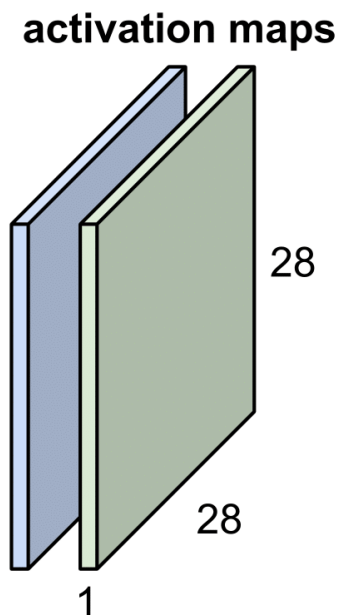
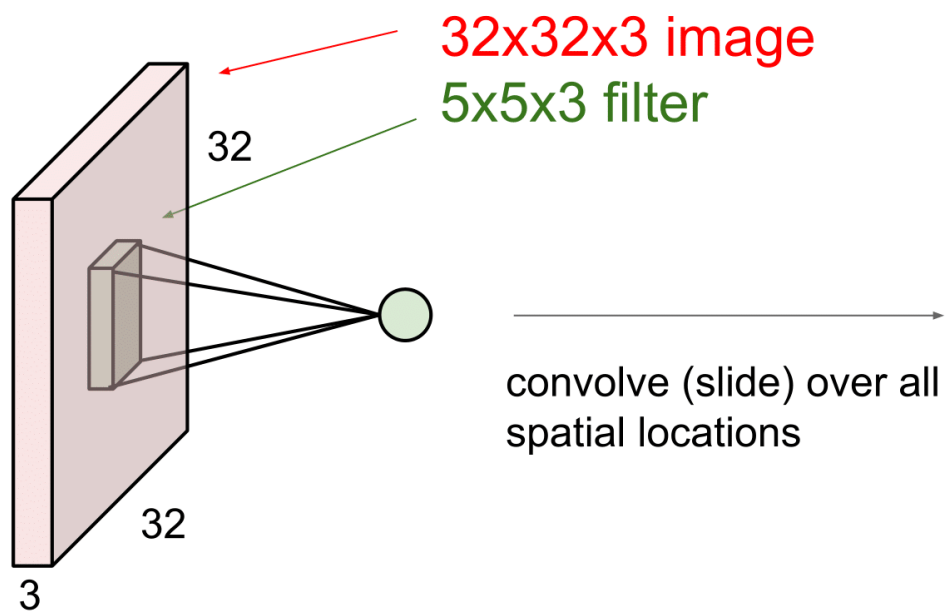
深度学习的结构突破



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- 循环神经网络 (RNN)
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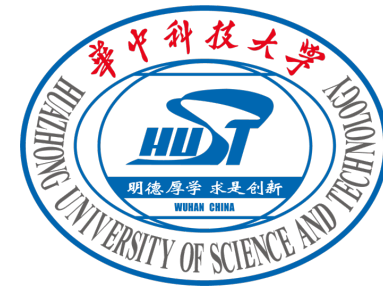
Convolution Layer

consider a second, **green** filter



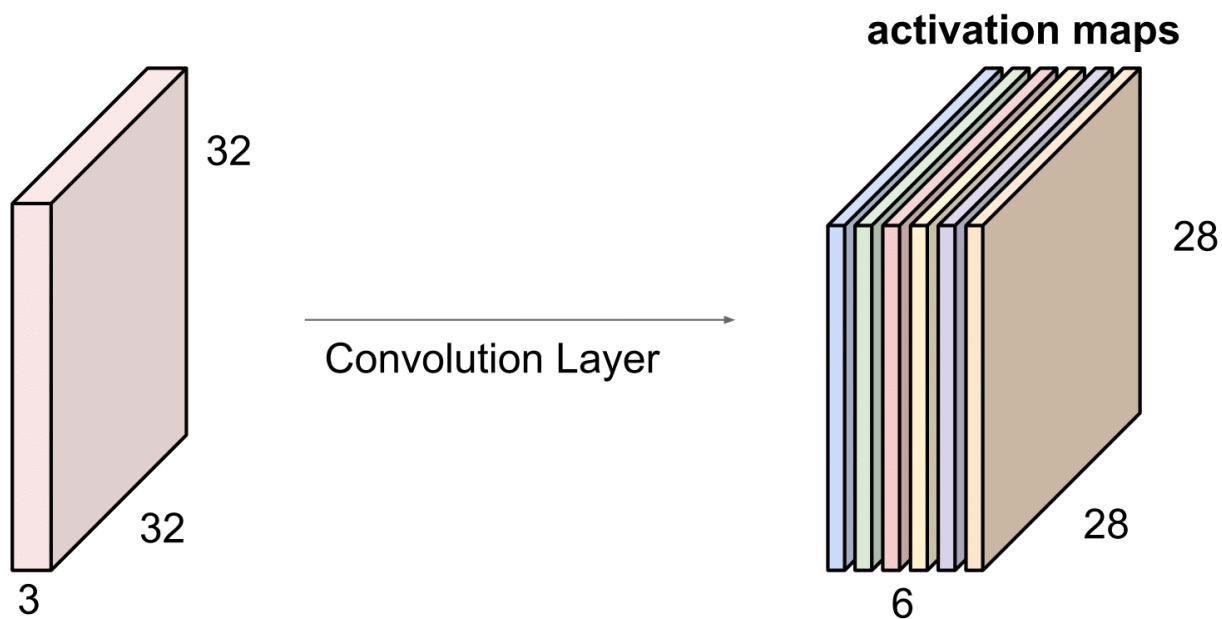


深度学习的结构突破

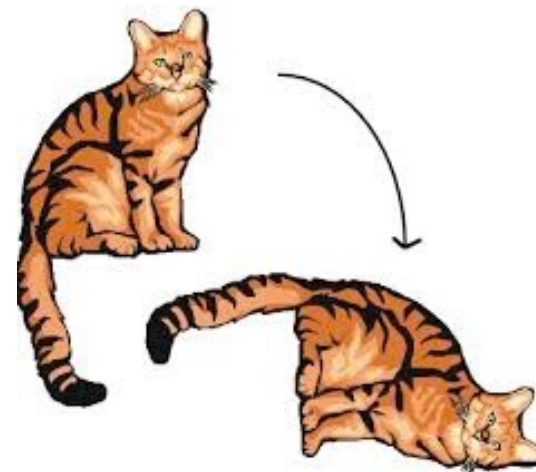


- 卷积神经网络 (CNN)
- 循环神经网络 (RNN)
- Transformer模型

For example, if we had 6 5x5 filters, we'll get 6 separate activation maps:

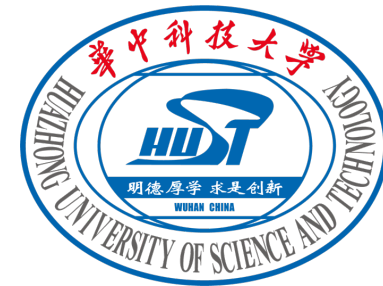


We stack these up to get a “new image” of size 28x28x6!

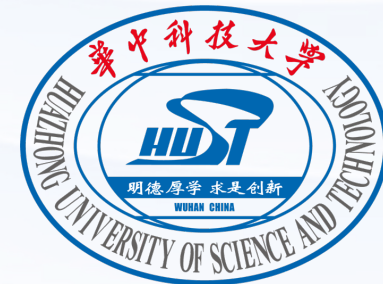




深度学习的结构突破



- 卷积神经网络 (CNN)
- 循环神经网络 (RNN)
- Transformer模型
 - 【【官方双语】大语言模型的简要解释】
https://www.bilibili.com/video/BV1xmA2eMEFF/?share_source=copy_web&vd_source=405ba20a8a5d0e3e737677cdbaa02da6
- 来点实例! <https://chat.deepseek.com/>

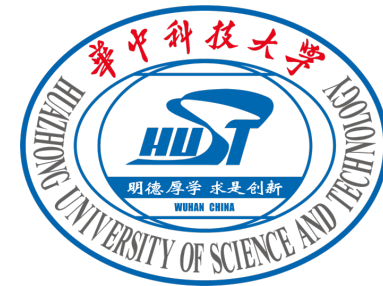


PART 05

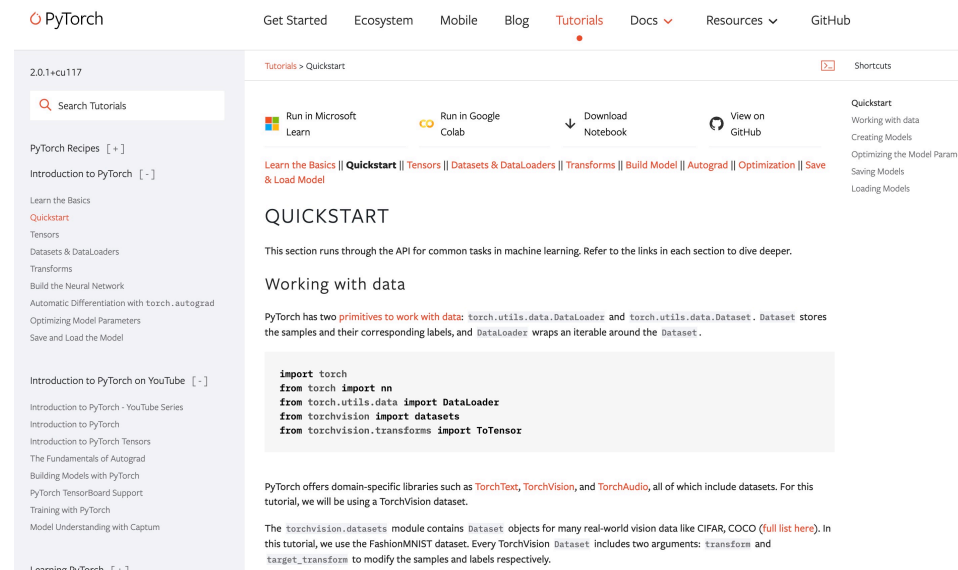
学习资源和结语



人工智能实践工具与平台：开源框架



- 深度学习领域有许多开源框架和工具，如 **PyTorch**、**Scikit-learn**、**TensorFlow**等，基于Python和相关科学计算库
 - 提供了丰富的API和预训练模型，方便开发者快速搭建和训练深度学习模型
 - <https://scikit-learn.org/stable/index.html>
 - <https://pytorch.org/> 尤其是对应的教程
https://pytorch.org/tutorials/beginner/basics/quickstart_tutorial.html
- 此外，还有许多开源的**数据集和工具库**，如UCI机器学习库、Kaggle数据集等，为机器学习和深度学习的实践提供了丰富的资源。





人工智能实践工具与平台：在线课程和参考书籍

Instructors



Fei-Fei Li

• 在线课程

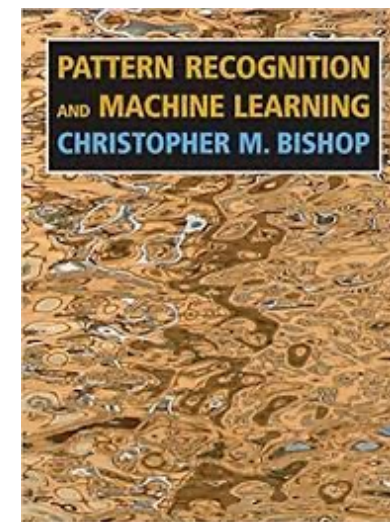
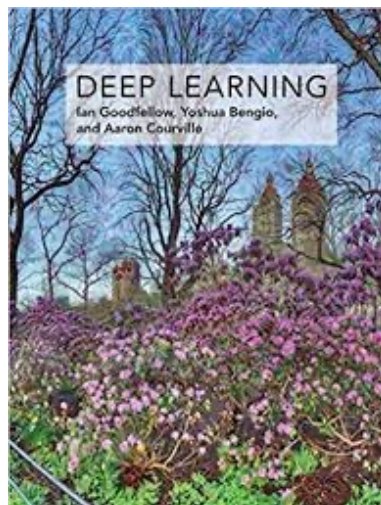
- Coursera的机器学习课程: <https://www.coursera.org/instructor/andrewng>
- 斯坦福大学的CS231n课程: CS231n: Deep Learning for Computer Vision (<http://cs231n.stanford.edu/>)



Andrew Ng
Co-founder, Coursera; Adjunct Professor, Stanford University; formerly head of Baidu AI Group/Google Brain
[斯坦福大学](#)
deeplearning.ai

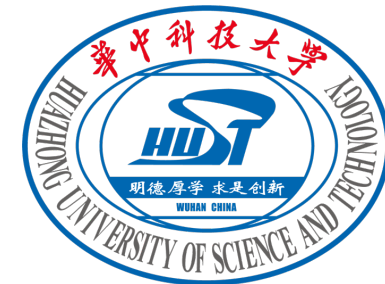
• 相关书籍

- 《深度学习》（花书）
- 《机器学习》（西瓜书）
- PRML





人工智能实践工具与平台：实践项目



- 参加实践项目和竞赛是提高人工智能实践能力的有效方式
 - **Kaggle**竞赛平台提供了各种机器学习和深度学习的竞赛项目，吸引了全球的开发者参与。
 - **Github**：还可以通过参与开源项目和社区活动
 - **Huggingface** <https://huggingface.co/huggingface>
 - <https://paperswithcode.com/>

Active	All	Entered	Sort by	Prize
19 active competitions			All Categories	Search competitions
	Passenger Screening Algorithm Challenge Improve the accuracy of the Department of Homeland Security's threat recognition algorithms Featured · a month to go · terrorism, image, object detection			\$1,500,000 400 teams
	Zillow Prize: Zillow's Home Value Prediction (Zestimate) Can you improve the algorithm that changed the world of real estate? Featured · 2 months to go · housing, real estate			\$1,200,000 3,780 teams
	Statoil/C-CORE Iceberg Classifier Challenge Ship or iceberg, can you decide from space? Featured · 2 months to go · weather, shipping, binary classification			\$50,000 1,100 teams
	Cdiscount's Image Classification Challenge Categorize e-commerce photos Featured · a month to go · multiclass classification			\$35,000 422 teams
	Corporación Favorita Grocery Sales Forecasting Can you accurately predict sales for a large grocery chain? Featured · 2 months to go · food and drink, tabular, regression, forecasting			\$30,000 628 teams
	TensorFlow Speech Recognition Challenge Can you build an algorithm that understands simple speech commands? Featured · 2 months to go			\$25,000 40 teams

Right Whale Recognition

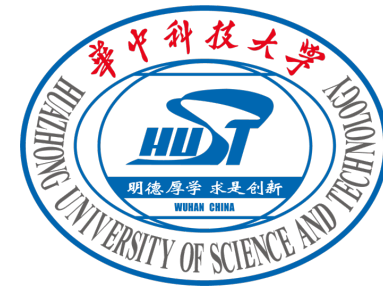
With fewer than 500 North Atlantic right whales left in the world's oceans, knowing the health and status of each whale is integral to the efforts of researchers working to protect the species from extinction.

Currently, only a handful of very experienced researchers can identify individual whales on sight while out on the water. For the majority of researchers, identifying individual whales takes time, making it difficult to effectively target whales for biological samples, acoustic recordings, and necessary health assessments.





相关领域的会议和期刊

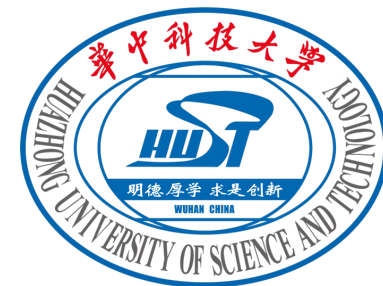


• 机器学习

- 顶级会议
 - ICML, NeurIPS, ICLR, COLT
 - AAAI, IJCAI
 - ECML, ACML
- 顶级期刊
 - Journal of Machine Learning Research (JMLR)
 - Machine Learning
 - Annals of Statistics

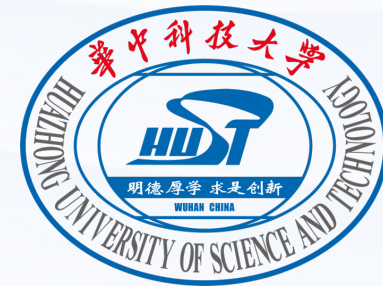
• 计算机视觉

- 顶级会议
 - CVPR, ICCV, ECCV, ICLR
 - BMVC, AAAI, IJCAI
 - ACCV, ICPR等
- 顶级期刊
 - IEEE Transactions on Pattern Analysis and Machine Intelligence (IEEE T-PAMI)
 - International Journal of Computer Vision (IJCV)



考核相关要求

- **考评内容（二选一）：**
 1. 机器学习人工智能技术在XX（问题、领域）中的应用
 2. 机器学习和人工智能的思考：过去、现状和未来
- **评价标准：**
 1. 应用选题：完整性和创新性（报告内容完成、逻辑清晰）
 2. 论文选题：报告完整、观点创新、思路深度、论证严谨
- **需要提交：**
 - 纸质报告（不少于5000字）：**课程结束后两周内**发至：
Zhenyu_liao@hust.edu.cn
 - 可附电子材料和附件，包括但不限于对应代码、项目效果展示多媒体等



感谢!

电子信息与通信学院
廖振宇